

DTSA 5740

Global Climate Policies and Analysis

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (59.6 / 100)	43.4 / 100
SPECIALIZATION Climate Change: Time Series Analysis		

43

CURRENT AVG SCORE
across 6 job roles

76

POTENTIAL SCORE
with recommended additions

+43

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course explores and critically analyzes historical and contemporary climate policies (e.g. Kyoto Protocol and the Paris Agreement). Political issues pertaining to energy sources, such as nuclear energy, will be reviewed. The course will focus on understanding key climate principles and terms surrounding policy development, specifically for low-income or developing countries/communities. Further, this course explores up-to-date technologies that are used in climate analysis. This course also introduces the Python programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	59.6	81.8	+22.2	MEDIUM
Machine Learning Engineer	53.1	83.2	+30.1	HIGH
ML Engineer	50.4	76.3	+25.9	MEDIUM
Data Scientist	37.5	73.8	+36.3	HIGH
Applied Scientist	31.4	74.3	+42.9	HIGH
MLOps Engineer	28.5	66.0	+37.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: LLMs & Generative AI — would broaden appeal to one additional role. Candidate terms: storage.

GAP KEYWORDS BY ROLE

AI Engineer

59.6 → 81.8 (+22.2 pts)

data models learning software model systems problems technology

Machine Learning Engineer

53.1 → 83.2 (+30.1 pts)

data predictive learning analytics tools science models model

ML Engineer

50.4 → 76.3 (+25.9 pts)

media data learning machine users systems models model

Data Scientist

37.5 → 73.8 (+36.3 pts)

toolkit data analytics different behaviors social problems approaches

Applied Scientist

31.4 → 74.3 (+42.9 pts)

computational research storage literature problems models data learning

MLOps Engineer

28.5 → 66.0 (+37.5 pts)

detection data model text systems learning machine media

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.